

WELL & OPERATIONAL METRICS

Well ID: OIL-BAGHJAN-4	Rig Name: BHEL F-3000 / Rig-14	Field: Baghjan (Assam Shelf)
Current Depth TVD: 2,465.0 m	Current Depth MD: 2,540.0 m	Kelly Bushing (KB): +108.5 m MSL
Active Horizon: Barail Formation	Present Mud Weight: 11.8 ppg	24-hr Progress: 48.0 m

24-HOUR OPERATIONAL SUMMARY & DRILLING NARRATIVE

DETAILED DOWNHOLE HAZARDS & NPT REMEDIATION RECORDS

Incident 1: Lost Circulation (Partial Loss)

Depth: 2425.0m TVD | NPT: 3.5 hrs | Severity: HIGH

Formation: Barail Formation (Upper Arenaceous Sandstone Member)

Root Cause: High differential pressure into depleted, micro-fractured reservoir sand causing fluid breakout.

Mitigation: Mixed and spotted 35 bbl LCM pill with nut-plug and mica. Soaked 2 hours and restored full circulation.

Incident 2: Gas Kick (Formation Influx)

Depth: 2460.0m TVD | NPT: 5.5 hrs | Severity: CRITICAL

Formation: Barail Formation (Overpressured Gas Sand Stringer)

Root Cause: Formation pore pressure (12.0 ppg equiv) exceeded active mud hydrostatic column resulting in gas influx.

Mitigation: Shut-in well on annular preventer. Recorded SIDPP and SICP, circulated out influx using Driller's Method with weighted kill mud.

Incident 3: Differential Sticking & Packoff

Depth: 2480.0m TVD | NPT: 4.0 hrs | Severity: HIGH

Formation: Barail Formation (Fissile Marine Shale Contact)

Root Cause: High hydrostatic overbalance causing drill collars to embed into thick mud cake during stationary survey.

Mitigation: Pumped 40 bbl Pipe-Lax surfactant soaking pill, applied 45 klbs overpull, and rotated slowly at 15 RPM to free drillstring.

DRILLING FLUID PROPERTIES & HYDRAULIC LOG

Mud Weight (Density):	11.8 - 12.2	ppg	11.2 - 12.8 ppg	Weighted up with barite post-kick
Plastic Viscosity (PV):	24	cP	18 - 30 cP	Normal hole cleaning condition
Yield Point (YP):	18	lb/100ft2	15 - 22 lb/100ft2	Maintained for Barail cuttings transport
API Fluid Loss:	4.5	ml/30min	< 5.0 ml	PAC-LV polymer filtration control
Equivalent Circulating Density:	12.45	ppg	Fracture Limit: 13.5 ppg	Safe drilling margin: 1.05 ppg

REPORT VALIDATION & CERTIFICATION SIGN-OFF

Prepared by:

Drilling Engineer / Tour Pusher
OIL Field Services, Baghjan Rig 14

Reviewed by:

Operations Geologist / Petrophysicist
Subsurface Directorate, OIL Duliajan

Approved by:

Superintendent Drilling (Operations)
Oil India Limited, Corporate HQ